Discord Math Server & Shark Math Program Mathematics Competition

2023 Round 2

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August 2023

Instructions and Information

1. DO NOT OPEN THE BOOKLET UNTIL YOUR CONTEST TIMER STARTS.
2. Teams can consist of up to 4 people. Each team should sign up before Round 1.
3. This is an 7-question, 2-day test, in which all of them are proof questions. Each question has a given weight (number of points) shown in the test beside the problem. The maximum number of points you can get is 50.
4. For all questions, please scan your papers and submit it to your assigned Discord channel and ping `mathemolympiad`.
5. No calculators, books, online resources, etc. are allowed, and no other tab aside from the test paper and the Discord contest channel should be opened. Only blank scratch paper, ruler, compass, and writing materials are allowed. If anyone is caught with illegal resources, they will be disqualified.
6. No person who participated or helped in the contest is allowed to discuss any aspect of it 48 hours from the contest date. If moderators or trial moderators catch anyone discussing during this time frame, they will automatically be disqualified.
7. Ties among two teams will be broken according to the team who scores higher in the last problem that their scores differ, then further ties will be broken according to their score in Round 1.

Good luck to all contestants and enjoy the contest!

Problems

1. **[5]** Let $n \geq 3$ be a positive integer. A country consists of n islands. To attract tourists, the government of the country decided to set up $n-1$ two-way tourism ship routes, in which each ship route connects two islands in the country, and one can travel between any two islands in the country only using the tourism ship routes. Now, you, being a government official, is tasked to give a *rating* of 1 or 2 to each of the $n-1$ tourism ship routes. The *attractiveness* of an island is the sum of all the ratings of the ship routes connecting to that island. A restriction is that the attractiveness of two islands directly connected by a tourism ship route has to be different. Show that, you can always assign a rating to all tourism ship routes satisfying the restriction, no matter how the tourism ship routes are set up.

2. **[7]** Find all triplets of positive integers (a, b, c) such that
$$\frac{\ln(ab+bc+ca)}{\ln(\operatorname{lcm}(a,b,c))}$$
is an integer.

3. **[7]** Let Γ be the circumcircle of an acute-angled $\triangle ABC$. A circle Ω is tangent to sides BC and AB at E and F respectively, and internally tangent to Γ at X. XE meets Γ again at $D \neq X$. Prove that $\dfrac{FA}{XA} = \dfrac{DC}{DX}$.

4. **[7]** Let $\{x_n\}_{n\geq 1}$ be a sequence of positive integers such that every positive integer appears in the sequence exactly once. Let $\{a_n\}_{n\geq 0}, \{b_n\}_{n\geq 0}$ be a sequence of real numbers such that $a_0 = 1, b_0 = 0$, and for any $n \in \mathbb{N}$,
$$a_n = 2a_{n-1}x_n - \frac{b_{n-1}}{x_n}$$
$$b_n = 2b_{n-1}x_n + \frac{a_{n-1}}{x_n}$$
Prove that for all $n \in \mathbb{N}$, $a_n > b_n$.

5. **[8]** Let n be a positive integer. Culver has to write a sequence of positive integers $a_1, a_2, ..., a_n$ such that for any positive integers l, r satisfying $1 \leq l \leq r \leq n$, we have
   - There exists a strict maximum x among $a_l, a_{l+1}, ..., a_r$, i.e. there exists an integer k between l and r inclusive such that $a_k = x$, and $a_i < x$ for any integer $i \neq k$ between l and r inclusive. Furthermore, x is a multiple of all of $a_l, a_{l+1}, ..., a_r$.
   - There exists a strict minimum y among $a_l, a_{l+1}, ..., a_r$, i.e. there exists an integer k between l and r inclusive such that $a_k = y$, and $a_i > y$ for any integer $i \neq k$ between l and r inclusive. Furthermore, y is a divisor of all of $a_l, a_{l+1}, ..., a_r$.

   Find the smallest possible value of the maximum integer among $a_1, a_2, ..., a_n$.

6. **[8]** Let p be an odd prime. Magma the rabbit is on the $\mathbb{F}_p^2$ land, which is a $p \times p$ grid with both coordinates numbered from 0 to $p-1$ going down to up and left to right. Furthermore, the 0-th row and the $(p-1)$-th row are connected, and the 0-th column and the $(p-1)$-th column are connected i.e. the cell below a cell on the 0-th row is the cell on the same column and on the $(p-1)$-th row, the cell above a cell on the $(p-1)$-th row is the cell on the same column and on the 0-th row, and similar applies for the columns.
   Hemu would like to teach Magma to perform some tricks. He firstly chooses an integer sequence $c_1, c_2, ..., c_n$. Magma then starts from $(0,0)$, and for the i-th move where $1 \leq i \leq n$, she chooses to jump c_i cells upwards, downwards, leftwards or rightwards. Hemu considers a cell *reachable* by the sequence $c_1, c_2, ..., c_n$ if there exists a way for Magma to jump such that she ends up in that cell after all n jumps. Now Hemu would like to impress Pyro, so he decides to let Pyro choose an integer k satisfying $0 \leq k < p$. Hemu then chooses an integer sequence $c_1, c_2, ..., c_n$ such that all cells of the form
$$\left(d^2 \bmod p, d^2k \bmod p\right)$$
   where $d \in \mathbb{Z}$ is reachable by $c_1, c_2, ..., c_n$. Hemu is smart, he always chooses the shortest possible sequence. But Pyro would like to make Magma feel exhausted, and hence would choose a k such that the shortest possible sequence that Hemu can choose is as long as possible. What will be the resulting length of the sequence?
   Note: $a \bmod p$ denotes the remainder of a divided by p.

7. **[8]** Let $n \geq 3$ be a positive integer. $A_1, A_2, ..., A_n$ are (possibly empty) subsets of the set $\left\{1, 2, ..., \frac{n(n-1)}{2}\right\}$. It is known that, for any positive integers i, j satisfying $1 \leq i, j \leq n$, the set $A_i \setminus A_j$ has cardinality at most $n-2$. Prove that the coefficient of the term $\prod_{i=1}^{\frac{n(n-1)}{2}} x_i$ in
$$\prod_{1 \leq i < j \leq n} \left(\sum_{k \in A_i} x_k - \sum_{k \in A_j} x_k \right)$$
   is zero.

Solutions

1. **[5]** Let $n \geq 3$ be a positive integer. A country consists of n islands. To attract tourists, the government of the country decided to set up $n - 1$ two-way tourism ship routes, in which each ship route connects two islands in the country, and one can travel between any two islands in the country only using the tourism ship routes. Now, you, being a government official, is tasked to give a *rating* of 1 or 2 to each of the $n - 1$ tourism ship routes. The *attractiveness* of an island is the sum of all the ratings of the ship routes connecting to that island. A restriction is that the attractiveness of two islands directly connected by a tourism ship route has to be different. Show that, you can always assign a rating to all tourism ship routes satisfying the restriction, no matter how the tourism ship routes are set up.

   Solution. The ship routes form a tree. Choose a degree 1 node as the root. For each node other than the root, assign the ratings of the edges such that a node with odd distance from the root has an odd attractiveness and a node with even distance from the root has an even attractiveness. This can always be done if the rating of an edge connecting a node and its parent is assigned after the rating of all other edges connecting to that node is assigned. This way the desired parity of attractiveness can be achieved for all non-root nodes. Therefore for any node that is not the root, the parity of the attractiveness of that node and the parity of the attractiveness of any of its children must be opposite i.e. the attractiveness of any that node must be not equal to the attractiveness of its children. Finally, for the root, since a root of degree 1 is picked, the set of edges that is connected to the root is a strict subset of the set of edges that is connected to the children of the root, hence the attractiveness of the root must be strictly smaller than the attractiveness of its children, hence their attractiveness is also not the same. $\square$

2. **[7]** Find all triplets of positive integers (a, b, c) such that

$$\frac{\ln(ab+bc+ca)}{\ln(\operatorname{lcm}(a,b,c))}$$

is an integer.
Answer: $(a, b, c) = (2n, 3n, 6n), (1, 2, 2)$ or $(3, 3, 3)$ and their permutations, where n is a positive integer.

Solution. The statement can be translated as there exists an integer k such that

$$ab + bc + ca = (\operatorname{lcm}(a, b, c))^k$$

Firstly notice that k must be positive, as a nonpositive k will yield the right hand side being at most 1, but the left hand side is at least $1 + 1 + 1 = 3$.
If $k = 1$, we have

$$ab + bc + ca = \operatorname{lcm}(a, b, c)$$

Since $ab, ca, \operatorname{lcm}(a, b, c)$ are all divisible by a, $a|bc$ i.e. $a^2|abc$. Similarly, $b^2|abc$ and $c^2|abc$. Hence $\operatorname{lcm}\left(a^2, b^2, c^2\right)|abc$. Pulling out the square yields $(\operatorname{lcm}(a, b, c))^2\,|abc$. However, $(\operatorname{lcm}(a, b, c))^2 = (ab+bc+ca)^2 > 2(ab)(bc) > abc$, so there are no solutions in the $k = 1$ case.
From here on WLOG assume $a \le b \le c$. If $k = 2$, we have

$$ab + bc + ca = (\operatorname{lcm}(a, b, c))^2$$

We can bound left hand side from above by $3c^2$. Hence $\operatorname{lcm}(a, b, c) \le \sqrt{3}c$. Since $\operatorname{lcm}(a, b, c)$ is a multiple of c, $\operatorname{lcm}(a, b, c) = c$, and consequently c are multiples of both a and b. Let $a' = \frac{c}{a}$ and $b' = \frac{c}{b}$, where a' and b' are positive integers. We divide both sides of the equation by ab and get

$$\begin{aligned} 1 + \frac{c}{a} + \frac{c}{b} &= \frac{c^2}{ab} \\ 1 + a' + b' &= a'b' \\ a'b' - a' - b' + 1 &= 2 \\ (a' - 1)(b' - 1) &= 2 \end{aligned}$$

Since $a \le b$, $a' \ge b'$ hence $(a' - 1, b' - 1) = (2, 1)$ which implies $(a', b') = (3, 2)$. Therefore $a : b : c = \frac{a}{c} : \frac{b}{c} : 1 = \frac{1}{3} : \frac{1}{2} : 1 = 2 : 3 : 6$. We can easily verify that $(a, b, c) = (2n, 3n, 6n)$ and its permutations work for n being a positive integer. If $k = 3$, we have

$$ab + bc + ca = (\operatorname{lcm}(a, b, c))^3$$

Notice that

$$3c^2 \ge ab + bc + ca = (\operatorname{lcm}(a, b, c))^3 \ge c^3$$

Hence $c \le 3$. If $c = 1$, then a and b must also be 1 but plugging in the equations shows that this does not work. If $c = 2$, since a and b are 1 or 2, $\operatorname{lcm}(a, b, c) = 2$ so we want $ab + 2a + 2b = ab + bc + ca = 2^3 = 8$ i.e. $(a + 2)(b + 2) = 12$. Since $3 \le a + 2 \le b + 2$, $a + 2 = 3$ and $b + 2 = 4$ and we get $(a, b, c) = (1, 2, 2)$. If $c = 3$, all the inequalities must become equality so $a = b = c = 3$. Hence the solutions in this case are $(a, b, c) = (1, 2, 2)$ or $(3, 3, 3)$ and their permutations, and it can be

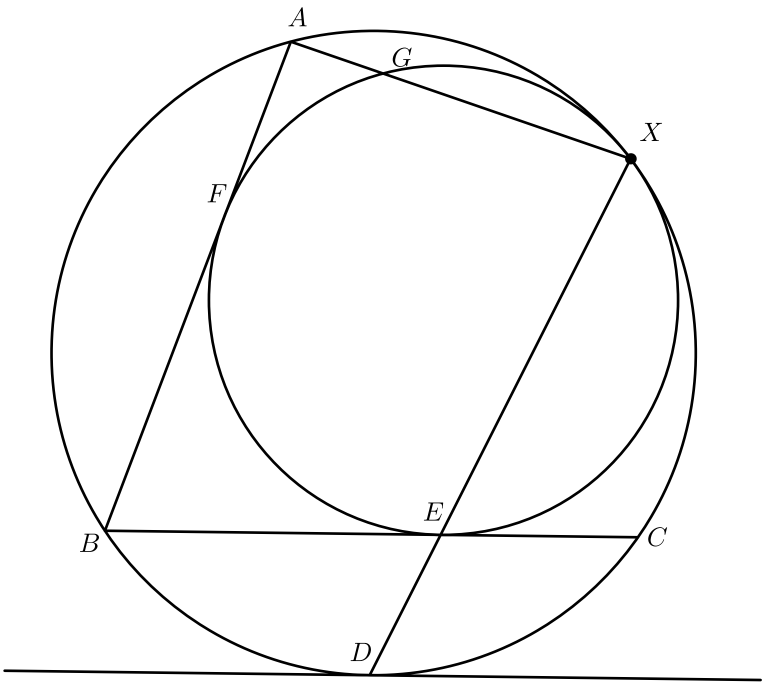
easily verified that these solutions work.
If $k = 4$, we can directly bound

$$3c^2 \geq ab + bc + ca = (\text{lcm}(a, b, c))^k \geq c^4$$

So $c^2 \geq 3$ which implies $c = 1$. Then a and b are also 1, but this obviously does not work. Hence there are no solutions in this case.
To conclude, all solutions are $(a, b, c) = (2n, 3n, 6n), (1, 2, 2)$ or $(3, 3, 3)$ and their permutations, where n is a positive integer. $\square$

3. **[7]** Let Γ be the circumcircle of an acute-angled $\triangle ABC$. A circle Ω is tangent to sides BC and AB at E and F respectively, and internally tangent to Γ at X. XE meets Γ again at $D \neq X$. Prove that $\dfrac{FA}{XA} = \dfrac{DC}{DX}$.

   Solution. We let $G \neq X$ be the other intersection point of AX and Ω.



A homothety centered at X that sends Ω to Γ also sends E to D. Therefore it also sends line BC to the line tangent to Γ at D. Hence BC is parallel to the line tangent to Γ at D and hence D is the midpoint of the minor arc BC. Consequently $\angle DCE = \angle DBC = \angle DXC$ i.e. $\triangle DEC \sim \triangle DCX$. Hence $\left(\dfrac{DC}{DX}\right)^2 = \dfrac{DE \cdot DX}{DX^2} = \dfrac{DE}{DX}$.

Also, we have $\left(\dfrac{FA}{XA}\right)^2 = \dfrac{GA \cdot XA}{XA^2} = \dfrac{GA}{XA}$.

Since the homothety centered at X that sends Ω to Γ also sends E to D and G to A, $\dfrac{DE}{DX} = \dfrac{GA}{XA}$ hence $\dfrac{FA}{XA} = \dfrac{DC}{DX}$ as desired. $\square$

4. **[7]** Let $\{x_n\}_{n\geq 1}$ be a sequence of positive integers such that every positive integer appears in the sequence exactly once. Let $\{a_n\}_{n\geq 0}, \{b_n\}_{n\geq 0}$ be a sequence of real numbers such that $a_0 = 1, b_0 = 0$, and for any $n \in \mathbb{N}$,

$$a_n = 2a_{n-1}x_n - \frac{b_{n-1}}{x_n}$$

$$b_n = 2b_{n-1}x_n + \frac{a_{n-1}}{x_n}$$

Prove that for all $n \in \mathbb{N}$, $a_n > b_n$.

Solution.

$$\begin{aligned}\frac{b_n}{a_n} &= \frac{2b_{n-1}x_n + \frac{a_{n-1}}{x_n}}{2a_{n-1}x_n - \frac{b_{n-1}}{x_n}} \\ &= \frac{\frac{b_{n-1}}{a_{n-1}} + \frac{1}{2x_n^2}}{1 - \frac{b_{n-1}}{a_{n-1}} \cdot \frac{1}{2x_n^2}} \\ \arctan\left(\frac{b_n}{a_n}\right) &= \arctan\left(\frac{b_{n-1}}{a_{n-1}}\right) + \arctan\left(\frac{1}{2x_n^2}\right)\end{aligned}$$

Since $\arctan\left(\frac{1}{2x_n^2}\right)$ is always positive, we must have

$$\begin{aligned}\arctan\left(\frac{b_n}{a_n}\right) &= \arctan\left(\frac{b_0}{a_0}\right) + \sum_{k=1}^{n} \arctan\left(\frac{1}{2x_k^2}\right) \\ &< \arctan\left(\frac{0}{1}\right) + \sum_{k=1}^{\infty} \arctan\left(\frac{1}{2k^2}\right) \\ &= 0 + \sum_{k=1}^{\infty} \arctan\left(\frac{\frac{1}{2k-1} - \frac{1}{2k+1}}{1 + \frac{1}{2k-1} \cdot + \frac{1}{2k+1}}\right) \\ &= \sum_{k=1}^{\infty}\left(\arctan\left(\frac{1}{2k-1}\right) - \arctan\left(\frac{1}{2k+1}\right)\right) \\ &= \arctan 1 \\ \frac{b_n}{a_n} &< 1 \\ a_n &> b_n\end{aligned}$$

As desired. $\square$

5. **[8]** Let n be a positive integer. Culver has to write a sequence of positive integers $a_1, a_2, ..., a_n$ such that for any positive integers l, r satisfying $1 \le l \le r \le n$, we have

   - There exists a strict maximum x among $a_l, a_{l+1}, ..., a_r$, i.e. there exists an integer k between l and r inclusive such that $a_k = x$, and $a_i < x$ for any integer $i \neq k$ between l and r inclusive. Furthermore, x is a multiple of all of $a_l, a_{l+1}, ..., a_r$.
   - There exists a strict minimum y among $a_l, a_{l+1}, ..., a_r$, i.e. there exists an integer k between l and r inclusive such that $a_k = y$, and $a_i > y$ for any integer $i \neq k$ between l and r inclusive. Furthermore, y is a divisor of all of $a_l, a_{l+1}, ..., a_r$.

   Find the smallest possible value of the maximum integer among $a_1, a_2, ..., a_n$.
   Answer: Smallest positive integer k such that $2^{\lfloor \frac{k}{2} \rfloor} + 2^{\lceil \frac{k}{2} \rceil} - 2 \ge n$.

   Solution. Let a positive integer s be in group $\lfloor \log_2 s \rfloor$.
   Claim: For any two positive integers in the sequence that has the same group, there exists an integer from a lower group number that is between the two positive integers, and an integer from a higher group number that is between the two positive integers.

   Proof. Let the two positive integers be at indices l and r and in group g, where $l < r$ Furthermore, let $b_1, b_2, ..., b_m$ be all indices where a_{b_i} is in group g for all $1 \le i \le m$ and $l < b_1 < b_2 < ... < b_m < r$. Consider the range $[l, b_1]$. Either $a_l = a_{b_i}$ or that a_l and a_{b_i} does not divide each other since the larger divided by the smaller of a_l and a_{b_i} must be less than 2, so they do not divide each other. If there does not exist a positive integer from a higher group number in $[l, r]$, then there does not exist a positive integer from a higher group number in $[l, b_1]$, hence the maximum in the range must be a_l or a_{b_i}. Since the maximum must be strict, a_l and a_{b_i} must not be equal, hence they do not divide each other. However, the strict maximum must be a multiple of every number in the range, a contradiction. Similar applies if there does not exist a positive integer from a lower group number in $[l, r]$. □

   As a corollary to the claim, the number of integers of the sequence that is in group g is at most one more than the number of integers of the sequence that has a group number less than g, and also at most one more than the number of integers of the sequence that is in a group that has a group number more than g, else we can find two positive integers in group g such that there does not exist an integer between them that has a lower group number, or there does not exist an integer between them that has a higher group number respectively.
   Then there are at most 1 group 0 number, $1 + 1 = 2$ group 1 numbers, $1 + 2 + 1 = 4$ group 2 numbers, and by simple induction there are at most 2^h group h numbers. Similarly, if the maximum group number among all of $a_1, a_2, ..., a_n$ is g, there are at most 1 group g number, $1 + 1 = 2$ group $g - 1$ numbers, $1 + 2 + 1 = 4$ group $g - 2$ numbers, and by simple induction there are at most 2^h group $g - h$ numbers. Taking the lower value to bound the maximum number of n if the maximum group number is g, we have the maximum value of n to be

$$2^{\lfloor \frac{k}{2} \rfloor} + 2^{\lceil \frac{k}{2} \rceil} - 2$$

   Indeed, we can construct a valid sequence by firstly placing $2^{v_2(i) + \lceil \frac{g}{2} \rceil}$ as the i-th number of the sequence for $1 \le i \le 2^{\lfloor \frac{g}{2} \rfloor + 1} - 1$, then for any integer s not equal to

$2^{\frac{g}{2}}$, place $\frac{2^g}{s}$ on the right of it, and at last take the first n integers of the sequence. Hence the answer is the smallest positive integer k such that $2^{\lfloor \frac{k}{2} \rfloor} + 2^{\lceil \frac{k}{2} \rceil} - 2 \geq n$. $\square$

6. **[8]** Let p be an odd prime. Magma the rabbit is on the $\mathbb{F}_p^2$ land, which is a $p \times p$ grid with both coordinates numbered from 0 to $p-1$ going down to up and left to right. Furthermore, the 0-th row and the $(p-1)$-th row are connected, and the 0-th column and the $(p-1)$-th column are connected i.e. the cell below a cell on the 0-th row is the cell on the same column and on the $(p-1)$-th row, the cell above a cell on the $(p-1)$-th row is the cell on the same column and on the 0-th row, and similar applies for the columns.
Hemu would like to teach Magma to perform some tricks. He firstly chooses an integer sequence $c_1, c_2, ..., c_n$. Magma then starts from $(0,0)$, and for the i-th move where $1 \leq i \leq n$, she chooses to jump c_i cells upwards, downwards, leftwards or rightwards. Hemu considers a cell *reachable* by the sequence $c_1, c_2, ..., c_n$ if there exists a way for Magma to jump such that she ends up in that cell after all n jumps. Now Hemu would like to impress Pyro, so he decides to let Pyro choose an integer k satisfying $0 \leq k < p$. Hemu then chooses an integer sequence $c_1, c_2, ..., c_n$ such that all cells of the form
$$\left(d^2 \bmod p, d^2 k \bmod p\right)$$
where $d \in \mathbb{Z}$ is reachable by $c_1, c_2, ..., c_n$. Hemu is smart, he always chooses the shortest possible sequence. But Pyro would like to make Magma feel exhausted, and hence would choose a k such that the shortest possible sequence that Hemu can choose is as long as possible. What will be the resulting length of the sequence?
Answer: $\lceil \log_2 p \rceil$

Solution: Do a transformation $(x,y) \to ((x+y) \bmod p, (x-y) \bmod p)$. Then the target cells will become $\left(d^2(k+1) \bmod p, d^2(k-1) \bmod p\right)$. A move by Magma then becomes changing each coordinate by c_i or $-c_i$ independently. Hence, for each integer m from 0 to $p-1$ that is representable as $d^2(k-1) \bmod p$ or $d^2(k+1) \bmod p$ for some $d \in \mathbb{Z}$, there must exists $a_1, a_2, ..., a_n \in \{-1, 1\}$ such that $a_1c_1 + a_2c_2 + ... + a_nc_n \equiv m \pmod p$.
Note that there must exist k such that one of $k-1$ and $k+1$ is a quadratic non-residue and the other being a non-zero quadratic residue modulo p, else we would have all intgers being a quadratic residue modulo p, which is absurd. Pyro can choose such k. Consider

$$1^2(k+1), 2^2(k+1), ..., \left(\frac{p-1}{2}\right)^2(k+1), 1^2(k-1), 2^2(k-1), ..., \left(\frac{p-1}{2}\right)^2(k-1)$$

Since $1^2, 2^2, ..., \left(\frac{p-1}{2}\right)^2$ gives distinct remainders modulo p,

$$1^2(k+1), 2^2(k+1), ..., \left(\frac{p-1}{2}\right)^2(k+1)$$

gives distinct remainders modulo p and

$$1^2(k-1), 2^2(k-1), ..., \left(\frac{p-1}{2}\right)^2(k-1)$$

gives distinct remainders modulo p. Furthermore, since one of $k+1$ and $k-1$ is a quadratic non-residue and the other bring a non-zero quadratic residue modulo p, we cannot have $d_1^2(k+1)$ and $d_2^2(k-1)$ being equal modulo p where $1 \leq d_1, d_2 \leq \frac{p-1}{2}$, since one of them is a quadratic non-residue and the other bring a non-zero quadratic residue modulo p. Therefore,

$$1^2(k+1), 2^2(k+1), ..., \left(\frac{p-1}{2}\right)^2(k+1), 1^2(k-1), 2^2(k-1), ..., \left(\frac{p-1}{2}\right)^2(k-1)$$

leaves remainders $1, 2, ..., p-1$ in some order when divided by p. Since $0^2(k-1) \equiv 0 \pmod{p}$, each integer m from 0 to $p-1$ is representable as $d^2(k-1) \bmod p$ or $d^2(k+1) \bmod p$ for some $d \in \mathbb{Z}$. On the other hand, since there are two choices for each a_i, the expression $a_1c_1 + a_2c_2 + ... + a_nc_n$ can take at most 2^n values. So we must have $2^n \geq p$ i.e. $n \geq \lceil \log_2 p \rceil$.
It suffices to construct a sequence $c_1, c_2, ..., c_n$ where $n = \lceil \log_2 p \rceil$. Indeed, we can let $c_i = 2^{i-1}$. Then $a_1c_1 + a_2c_2 + ... + a_nc_n$ can be any odd integer from $-2^n + 1$ to $2^n - 1$. Since there are 2^n odd integers from $-2^n + 1$ to $2^n - 1$, which is at least p, taking any p consecutive odd numbers from it yields a complete residue class modulo p so we are done. $\square$

7. **[8]** Let $n \geq 3$ be a positive integer. $A_1, A_2, ..., A_n$ are (possibly empty) subsets of the set $\left\{1, 2, ..., \frac{n(n-1)}{2}\right\}$. It is known that, for any positive integers i, j satisfying $1 \leq i, j \leq n$, the set $A_i \setminus A_j$ has cardinality at most $n-2$. Prove that the coefficient of the term $\prod_{i=1}^{\frac{n(n-1)}{2}} x_i$ in

$$\prod_{1 \leq i < j \leq n} \left(\sum_{k \in A_i} x_k - \sum_{k \in A_j} x_k \right)$$

is zero.

Solution. We claim that if $x_1, x_2, ..., x_{\frac{n(n-1)}{2}} \in \{0, 1\}$, the value of

$$\prod_{1 \leq i < j \leq n} \left(\sum_{k \in A_i} x_k - \sum_{k \in A_j} x_k \right)$$

is zero.

Proof. Indeed we can let $s_i = \sum_{k \in A_i} x_k$. Then s_i must be an integer. We can see for any positive integers i, j satisfying $1 \leq i, j \leq n$,

$$s_i - s_j = \sum_{k \in A_i} x_k - \sum_{k \in A_j} x_k = \sum_{k \in A_i \setminus A_j} x_k - \sum_{k \in A_j \setminus A_i} x_k \leq |A_i \setminus A_j| - 0 = n - 2$$

So we must have $max(s_1, s_2, ..., s_n) - min(s_1, s_2, ..., s_n) \leq n - 2$. Since $s_1, s_2, ..., s_n$ are integers, the number of distinct values among $s_1, s_2, ..., s_n$ is at most $n - 2 + 1 = n - 1$. Hence there must exist $1 \leq i < j \leq n$ such that $s_i = s_j$. Since the product of the polynomial includes $\sum_{k \in A_i} x_k - \sum_{k \in A_j} x_k$ which is $s_i - s_j$, so the polynomial would become 0 and that completes the proof of the claim. $\square$

Since the degree of the polynomial is $\frac{n(n-1)}{2}$, together with the claim, we can apply the contrapositive of Combinatorial Nullstellensatz to conclude the coefficient of the term $\prod_{i=1}^{\frac{n(n-1)}{2}} x_i$ in the polynomial is zero. $\square$